

HOMEWORK 2

1. Prove that every finite integral domain is a field. Deduce that every prime ideal of a finite ring is maximal.
2. Prove that every ring is isomorphic to the quotient ring $\mathbb{Z}\langle X \rangle / I$ for a set X and an ideal $I \subset \mathbb{Z}\langle X \rangle$. (Here $\mathbb{Z}\langle X \rangle$ is the polynomial ring of noncommuting set of variables X over $\mathbb{Z}$.)
3. Prove that the category *Rings* admits coproducts.
4. Let R be a commutative ring such that every $a \in R$ is idempotent (i.e., $a^2 = a$). Prove that every prime ideal in R is maximal.
5. Prove that every nonzero commutative ring has a minimal (by inclusion) prime ideal.
6. The *spectrum* $\text{Spec}(R)$ of a commutative ring R is the set of all prime ideals of R . Determine the spectrum of the subring $R \subset \mathbb{Q}$ of all fractions $\frac{a}{b}$ with b an odd integer.
7. For two commutative rings R and S construct a bijection between $\text{Spec}(R \times S)$ and the disjoint union of $\text{Spec } R$ and $\text{Spec } S$.
8. Let R be a commutative ring and $I \subset R$ an ideal. Construct a bijection between $\text{Spec}(R/I)$ and the subset in $\text{Spec } R$ of all prime ideals containing I .
9. Let R be a commutative ring and $S \subset R$ a *multiplicative subset*, i.e., a subset containing 1 and closed under multiplication. Define a relation $\sim$ on the set $R \times S$: $(r_1, s_1) \sim (r_2, s_2)$ if there is an element $s \in S$ such that $s(r_1 s_2 - r_2 s_1) = 0$.
 - a) Prove that $\sim$ is an equivalence relation. Write $\frac{r}{s}$ for the equivalence class of (r, s) and $S^{-1}R$ for the set of equivalence classes.
 - b) Prove that the binary operations $\frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}$ and $\frac{r_1}{s_1} \cdot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}$ are well defined and make $S^{-1}R$ a commutative ring.
 - c) Prove that the map $f : R \rightarrow S^{-1}R$, taking r to $\frac{r}{1}$ is a ring homomorphism.
10. Let R be a commutative ring and $S \subset R$ a multiplicative subset. Let $g : R \rightarrow T$ be a homomorphism of commutative rings such that $g(S) \subset T^\times$. Prove that there is a unique ring homomorphism $h : S^{-1}R \rightarrow T$ such that $g = h \circ f$, where $f : R \rightarrow S^{-1}R$ is the homomorphism from Problem 9c.